



EXECUTIVE FINANCIAL DASHBOARD TECHNICAL DOCUMENT

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Overview

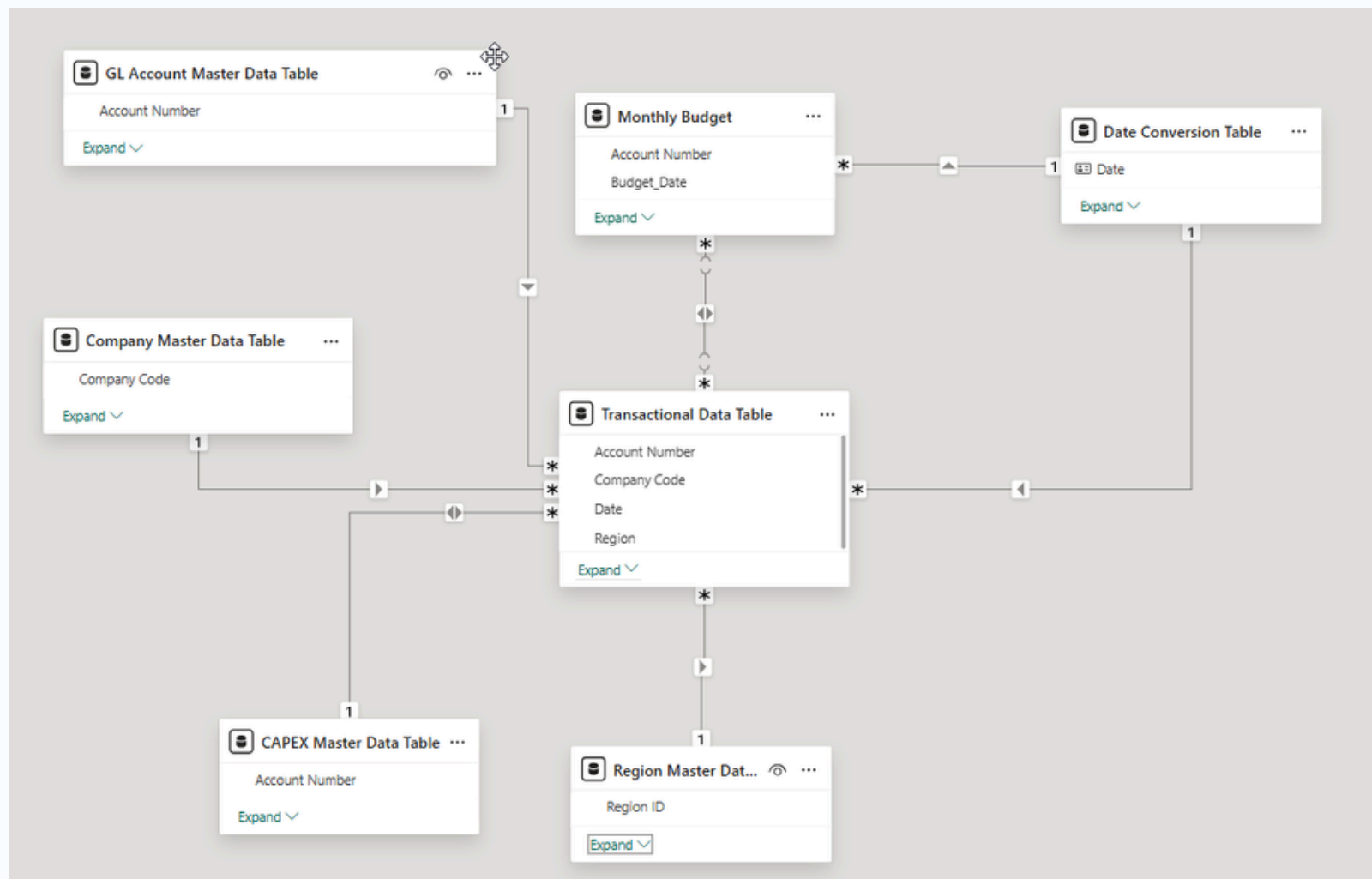
This technical documentation presents an enterprise-grade financial analytics solution engineered for holding-level executives, CFOs, and FP&A leaders. Designed to consolidate multi-entity corporate financial data, this architecture bridges operational ledger transactions with strategic executive Decision Support Systems (DSS).

The dashboard suite integrates three core analytical pillars:

- **Budget vs. Actual Variance Overview:** Granular evaluation of operational revenue, expense, and net income performance against fiscal targets.
- **Corporate Performance & CAPEX Allocation:** Consolidated view of EBITDA performance, entity-level contribution, and strategic capital expenditure distribution.
- **Trade Cycle & Working Capital Analytics:** Deep-dive liquidity assessment covering DSO, DPO, DIO, Cash Conversion Cycle (CCC), and balance sheet aging buckets.

Data Modeling

The data layer employs an enterprise Star Schema engineered to seamlessly bridge fine-grained daily transaction postings with monthly aggregated budget targets. By leveraging optimized bridge relationships and strictly controlled cross-filter propagation, the architecture delivers deterministic multi-grain consolidation while preserving high-speed DAX evaluation.



Manage relationships

+ New relationship

⚡ Autodetect

Edit

Delete

Filter

☐	From: table (column) ↑	Relationship	To: table (column)	Status	
☐	Monthly Budget (Budget_Date)	* ← 1	Date Conversion Table (Date)	Active	...
☐	Transactional Data Table (Acco...	* ↔ *	Monthly Budget (Account Nu...	Active	...
☐	Transactional Data Table (Acco...	* ← 1	GL Account Master Data Table ...	Active	...
☐	Transactional Data Table (Acco...	* ← 1	CAPEX Master Data Table (Acc...	Active	...
☐	Transactional Data Table (Com...	* ← 1	Company Master Data Table (...)	Active	...
☐	Transactional Data Table (Date)	* ← 1	Date Conversion Table (Date)	Active	...
☐	Transactional Data Table (Regi...	* ← 1	Region Master Data Table (Reg...	Active	...

Schema Breakdown & Table Taxonomy

- **Fact Tables:**

- Transactional Data Table : Captures daily transactional postings including actual financial movements (Actual_Amount), entity keys, GL account numbers, and posting dates.
- Monthly Budget : Houses monthly budget targets (Realistic_Budget) across GL account numbers for FY2025.

- **Dimension Tables:**

- Company Master Data Table: Lookup dimension defining corporate entities (Centidel Group, Cogilith Group, Skyndu Solutions).
- GL Account Master Data Table: Chart of Accounts mapping GL codes to higher-level P&L and Balance Sheet line items (Revenue, COGS, OPEX, Departments).
- CAPEX Master Data Table: Maps capital expenditure GL lines to strategic asset categories (Infrastructure, Equipment, IT Investment, Mobility Assets).
- Region Master Data Table: Regional operational mapping (Central, East, West).
- Date Conversion Table: Central date dimension providing complete time-intelligence capabilities.
- User_Security: Security mapping table driving dynamic user access rights based on USERPRINCIPALNAME().

Multi-Dimensional Dynamic Row-Level Security (RLS)

In a multi-subsidiary corporate environment, data governance is paramount. Executive leaders (CEOs/CFOs) require aggregated cross-entity access, while regional managers or departmental heads must only inspect data pertinent to their jurisdiction. Rather than creating static, hardcoded roles for every user, this architecture implements Dynamic Security using a single role (Dynamic_Security) driven by USERPRINCIPALNAME().

Technical Implementation Mechanics

- **Security Mapping Table (User_Security):** Maintains user email identifiers alongside their permitted Company Code and Allowed Department assignments.
- **Dimension-Level Isolation:** RLS expressions are applied directly to the primary lookup tables (Company Master Data Table and GL Accounts Master Data Table). This guarantees that restricted users cannot see unpermitted entities even in slicer options or axis legends.
- **Handling "ALL" Privileges:** Evaluates whether a user holds explicit wildcard access ("ALL"), granting full holding-level visibility (e.g., C-Level executives), or restricts filter context to specific company/department combinations.

Multi-Dimensional Dynamic Row-Level Security (RLS)

Applied DAX Expression

Manage security roles

Create new security roles and use filters to define row-level data restrictions.

Roles

- + New
- Dynamic_Security

Select tables

- _All Measures
- Aging Buckets
- CAPEX Master ...
- CAPEX Master ...
- Company Mas...
- Company Mas...
- Date Conversi...
- GL Account M...

Filter data

```
1 VAR CurrentUser = USERPRINCIPALNAME()
2
3 VAR UserCompanies =
4     CALCULATETABLE(
5         VALUES(User_Security[Company Code]),
6         User_Security[UserEmail] = CurrentUser
7     )
8
9 RETURN
10 "ALL" IN UserCompanies || [Company Code] IN UserCompanies
```

Switch to default editor

Applied DAX Expression

Manage security roles

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Roles

- + New
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Select tables

- _All Measures
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- CAPEX Master ...
- Company Mas...
- Company Mas...
- Date Conversi...
- GL Account M...

Filter data

```
1 VAR CurrentUser = USERPRINCIPALNAME()
2
3 VAR UserDepts =
4     CALCULATETABLE(
5         VALUES(User_Security[Allowed Department]),
6         User_Security[UserEmail] = CurrentUser
7     )
8
9 VAR UserCompanies =
10    CALCULATETABLE(
11        VALUES(User_Security[Company Code]),
12        User_Security[UserEmail] = CurrentUser
13    )
14
```

Switch to default editor

Row-Level Security (RLS)

View as roles

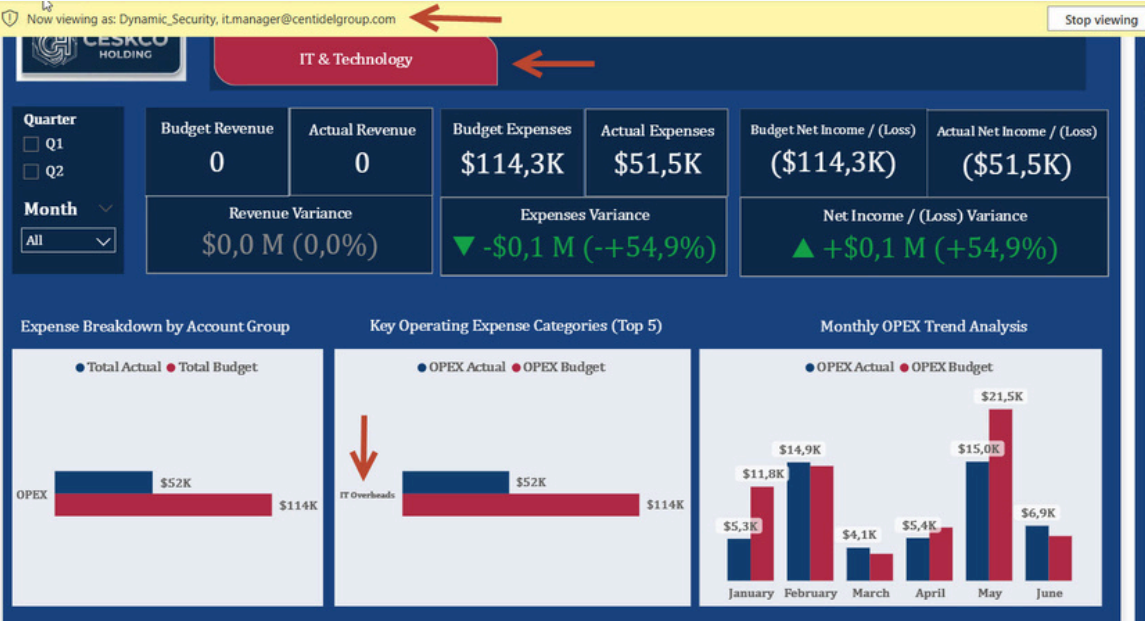
☐ None

☒ Other user

it.manager@centidelgroup.c

☒ Dynamic_Security

Role : IT Manager



View as roles

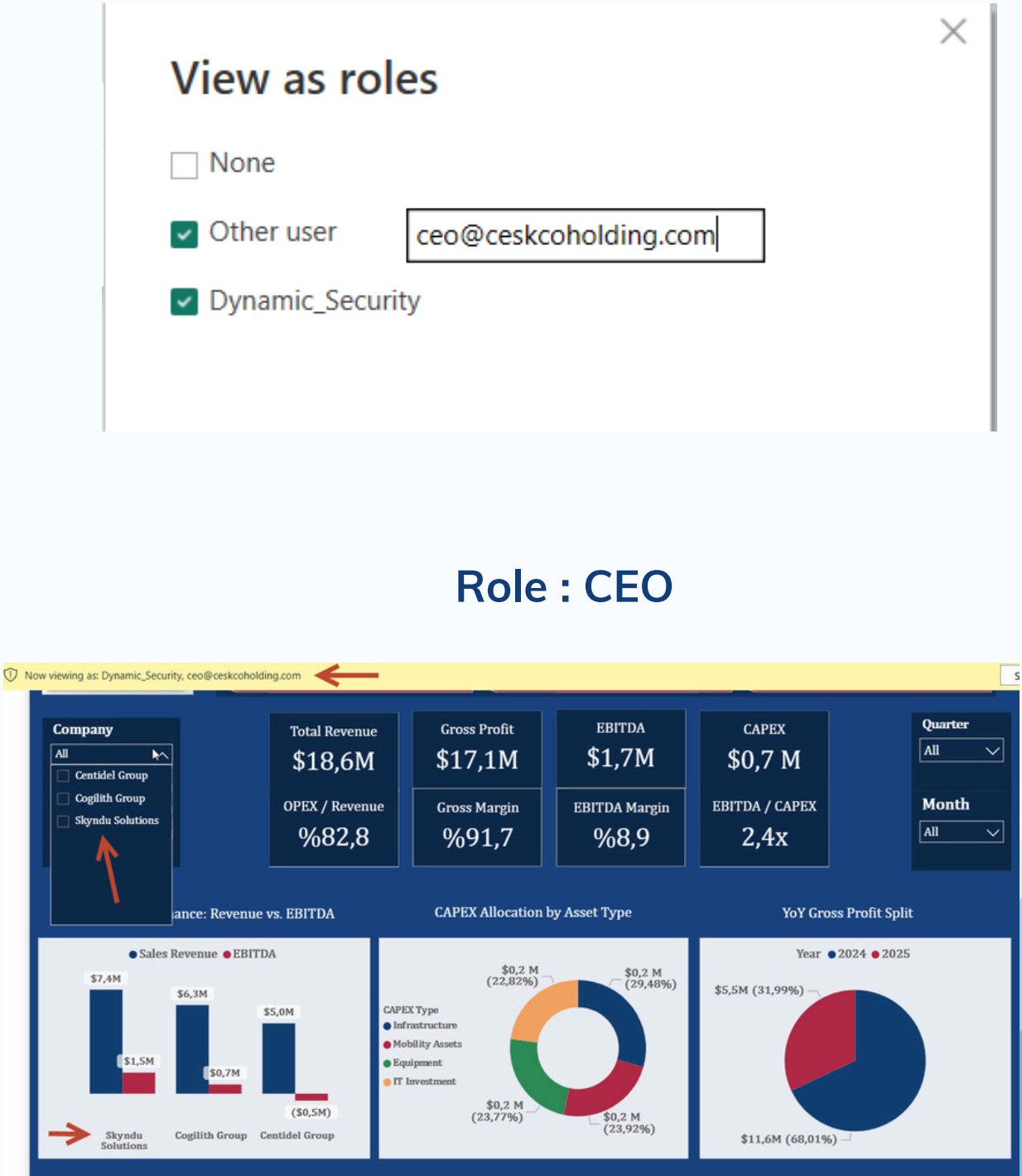
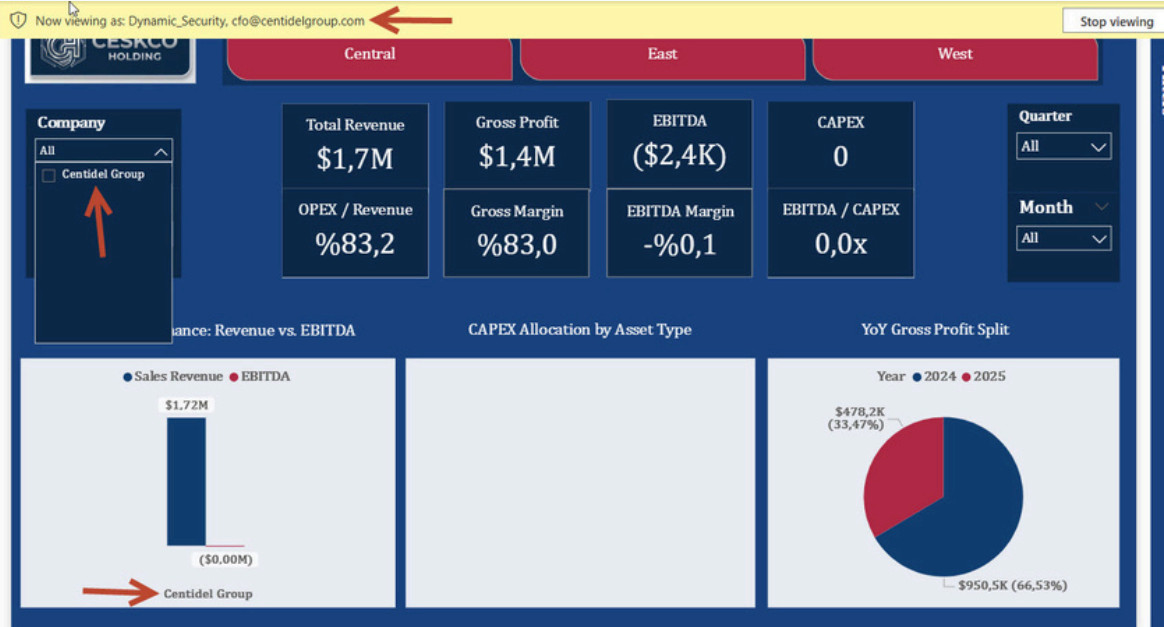
☐ None

☒ Other user

cfo@centidelgroup.com

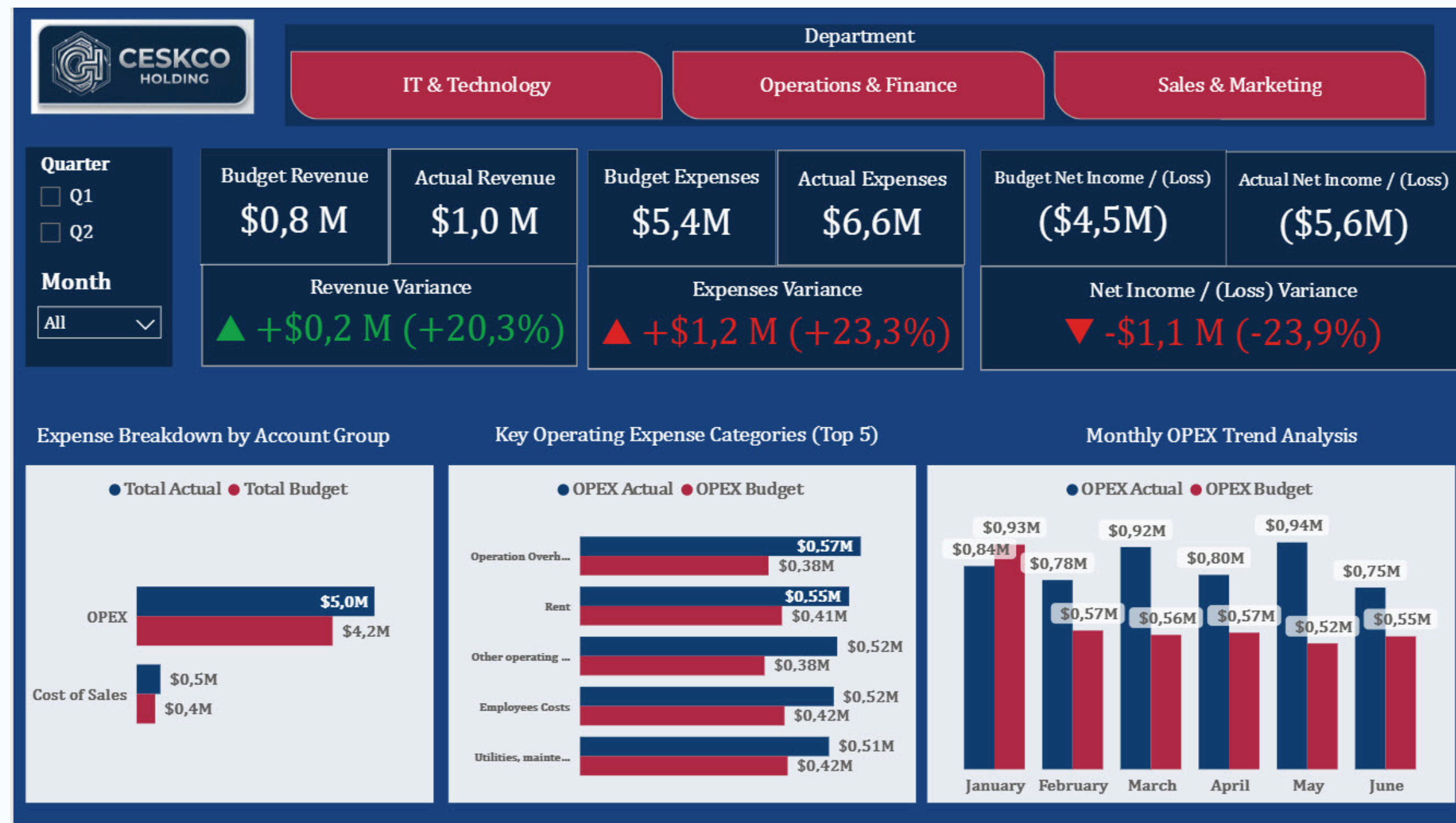
☒ Dynamic_Security

Role : CFO



Budget vs. Actual Overview

The primary objective of FP&A is measuring operational variance against predefined financial targets. This dashboard compares H1 2025 daily actuals against monthly budget plans, enabling management to identify expense overruns and revenue shortfalls early in the fiscal year.

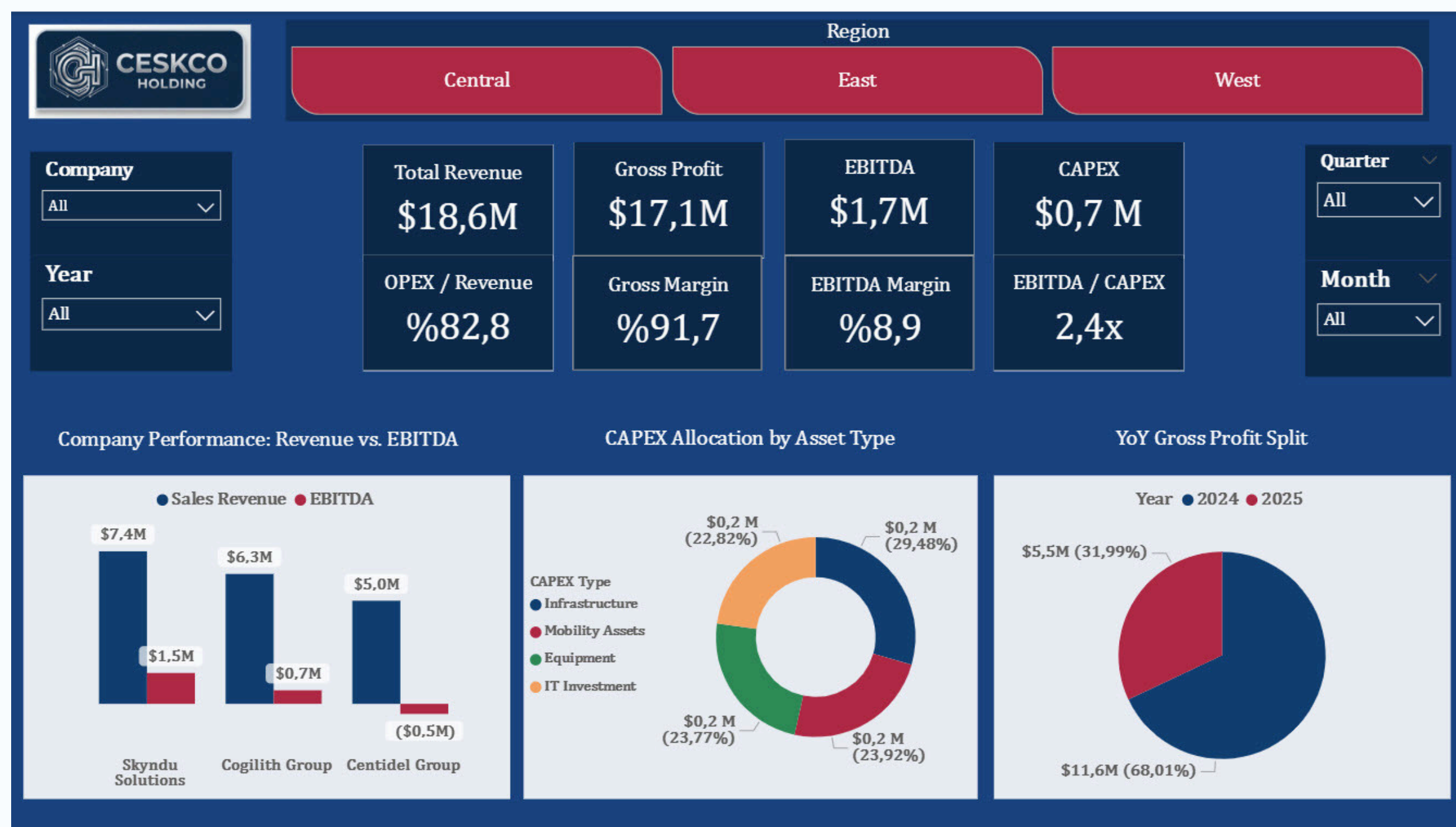


KPI Architecture & Financial Rationale

- **Budget vs. Actual Revenue (\$0.8M Budget vs. \$1.0M Actual):**
 - Formula: Actual Revenue - Budget Revenue
 - Business Impact: Indicates top-line commercial health. A positive variance (+\$0.2M / +20.3%) demonstrates strong market execution exceeding plan expectations.
- **Expenses Variance (\$5.4M Budget vs. \$6.6M Actual):**
 - Formula: Actual Expenses - Budget Expenses
 - Business Impact: Highlights operational cost control. An unfavorable variance (+\$1.2M / +23.3%) alerts executive leadership to unexpected operational overheads or cost inflation.
- **Net Income / (Loss) Variance (-\$4.5M Budget vs. -\$5.6M Actual):**
 - Formula: Actual Net Income - Budget Net Income
 - Business Impact: Measures bottom-line erosion (-\$1.1M / -23.9%). Demonstrates how operational expense growth outpaced top-line revenue gains.
- **Expense Breakdown by Account Group & Key OPEX Categories:**
 - Business Value: Allows managers to pinpoint specific cost centers driving negative bottom-line variances.

Corporate Performance & CAPEX Allocation

Consolidating multi-subsidary financials requires evaluating earnings quality alongside long-term capital deployments. This dashboard focuses on EBITDA generation, operating margins, and Capital Expenditure (CAPEX) allocation across the group's asset portfolio.





KPI Architecture & Financial Rationale

Total Revenue (\$18.6M) & OPEX / Revenue (%82.8):

- Financial Rationale: Total Revenue represents aggregate top-line corporate earnings, while the OPEX / Revenue ratio measures the proportion of total revenues consumed by ongoing operational overhead.
- Business Value: An 82.8% operational cost intensity highlights significant structural expense overhead across holding entities, indicating that cost-containment measures and operational optimization are necessary to preserve bottom-line margins.

Gross Profit (\$17.1M) & Gross Margin (%91.7):

- Financial Rationale: Gross Profit isolates core sales profitability after direct Cost of Goods Sold (COGS), with Gross Margin evaluating the efficiency of raw production/service delivery relative to total revenue.
- Business Value: An exceptional 91.7% Gross Margin proves extremely strong pricing power and high gross value delivery across business units, confirming that operating margin compression occurs primarily at the OPEX level rather than direct product costs.

EBITDA (\$1.7M) & EBITDA Margin (%8.9):

- Financial Rationale: Earnings Before Interest, Taxes, Depreciation, and Amortization isolates core operational profitability from capital structure and non-cash accounting decisions.
- Business Value: Serving as the primary metric for operational efficiency, an 8.9% EBITDA margin highlights an operational bottleneck where high Gross Profit (\$17.1M) is heavily absorbed by operational expenses, revealing clear operating leverage capacity across holding entities.

CAPEX (\$0.7M) & EBITDA / CAPEX Ratio (2.4x):

- Financial Rationale: Measures the enterprise's ability to cover its physical asset investments using internally generated operating cash flow.
- Business Value: A 2.4x coverage ratio confirms that operating cash flows fund ongoing asset expansion without over-leveraging corporate credit lines or relying on external capital injections.

Company Performance (Revenue vs. EBITDA):

- Financial Rationale: Evaluates revenue scale against core operating profitability across individual group subsidiaries to identify underlying margin drivers and cash drains.
- Business Value: Skyndu Solutions dominates group EBITDA contribution (\$1.5M on \$7.4M revenue), while Centidel Group exhibits negative operational performance (-\$0.5M EBITDA on \$5.0M revenue), identifying Centidel as the primary operational drag on consolidated holding margins.

CAPEX Allocation by Asset Type:

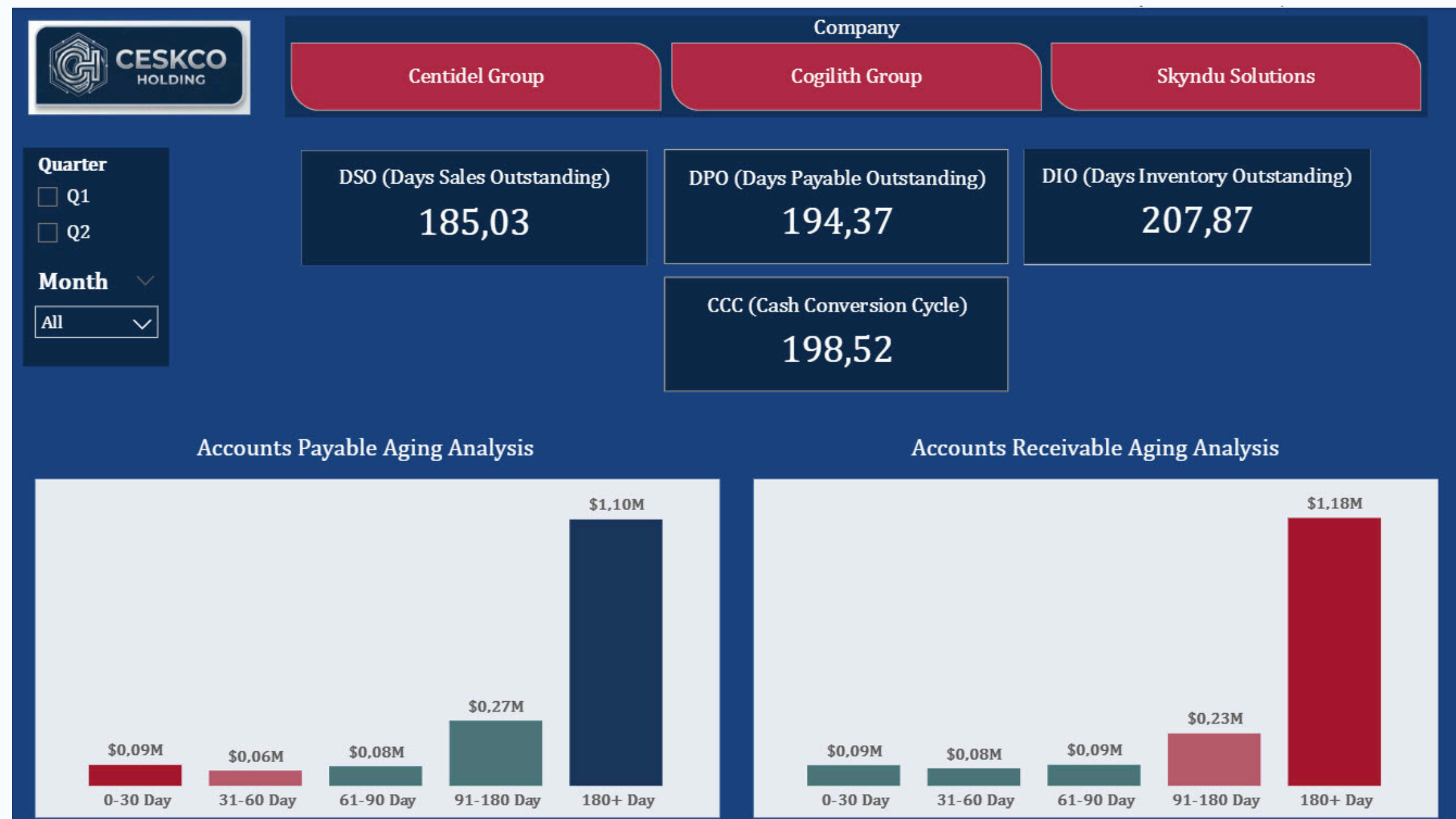
- Financial Rationale: Tracks capital allocation strategies across physical and technological assets to evaluate long-term infrastructure investment balance.
- Business Value: Capital expenditure is evenly distributed across core asset categories—led by Infrastructure (29.5%) and Mobility Assets (23.9%)—reflecting balanced asset maintenance without capital over-concentration in a single investment class.

YoY Gross Profit Split (2024 vs. 2025):

- Financial Rationale: Compares historical gross profit contribution across consecutive operating years to measure multi-year margin trajectory.
- 

Trade Cycle & Working Capital

Working Capital Management is the lifeblood of corporate liquidity. The Trade Cycle (Cash Conversion Cycle - CCC) quantifies the exact time (in days) required for an enterprise to convert its investments in inventory and operational resources into actual cash inflows from customer sales.



KPI Architecture & Financial Rationale

- **DSO (Days Sales Outstanding):**
 - Formula: $(\text{Accounts Receivable} / \text{Total Sales}) * \text{Period Days}$
 - Financial Impact: Measures average collection time. High DSO signals credit collection inefficiencies and increased bad debt risk.
- **DPO (Days Payable Outstanding):**
 - Formula: $(\text{Accounts Payable} / \text{Total Cost of Sales}) * \text{Period Days}$
 - Financial Impact: Reflects the average timeframe taken to pay suppliers. Extending DPO preserves cash reserves, but excessive delays may strain vendor relationships.
- **DIO (Days Inventory Outstanding):**
 - Formula: $(\text{Average Inventory} / \text{Total Cost of Sales}) * \text{Period Days}$
 - Financial Impact: Tracks inventory holding duration before conversion to sales. High DIO indicates working capital tied up in stagnant stock.
- **CCC (Cash Conversion Cycle):**
 - Formula: $\text{DSO} + \text{DIO} - \text{DPO}$
 - Calculated: $185.03 + 207.87 - 194.37 = 198.53 \text{ Days}$
 - Strategic Importance: Represents net liquidity delay. A shorter CCC reflects a highly efficient working capital engine.
- **Accounts Payable & Receivable Aging Buckets:**
 - Visual Mechanics: Categorizes outstanding receivables and payables into chronological risk bands (0-30 Days, 31-60 Days, 61-90 Days, 91-180 Days, 180+ Days).
 - Business Value: Enables immediate liquidity risk mitigation by highlighting severely overdue receivables (>180 Days).

Power Query M Formulas

```
= Table.AddColumn("#Changed Type", "Department", each if List.Contains({"Sales", "Cost of Sales",  
"Travel", "Other Income", "Marketing & Sales Expenses"}, Text.Trim([GL Category])) then "Sales &  
Marketing" else if List.Contains({"IT Overheads", "CAPEX - Capitalized Software", "CAPEX - Infrastructure"},  
Text.Trim([GL Category])) then "IT & Technology" else "Operations & Finance")
```

```
= Table.AddColumn("#Changed Type1", "OPEX", each if List.Contains({"Employees Costs", "Rent", "IT  
Overheads", "Travel", "Utilities, maintenance and supplies", "General & Admin", "Operation Overheads",  
"Insurance", "Other operating costs", "Marketing & Sales Expenses"}, Text.Trim([GL Category]))  
then "OPEX"  
else [GL Category])
```

Key DAX Calculations

```
Date Conversion Table =  
VAR MinYear = 2024  
VAR MaxYear = 2025  
RETURN  
ADDCOLUMNS (  
    CALENDAR ( DATE ( MinYear, 1, 1 ), DATE ( MaxYear, 12, 31 ) ),  
    "Year", YEAR ( [Date] ),  
    "Quarter_Num", QUARTER ( [Date] ),  
    "Quarter", "Q" & QUARTER ( [Date] ),  
    "Month_Num", MONTH ( [Date] ),  
    "Month_Name", FORMAT ( [Date], "mmmm", "en-US" ),  
    "Month_Short", FORMAT ( [Date], "mmm" ),  
    "Year_Month", FORMAT ( [Date], "yyyy-mm" ),  
    "Year_Month_Key", YEAR ( [Date] ) * 100 + MONTH ( [Date] ),  
    "Quarter_Year", FORMAT ( [Date], "yyyy" ) & "-Q" & QUARTER ( [Date] ),  
    "Day_of_Month", DAY ( [Date] ),  
    "Day_Name", FORMAT ( [Date], "dddd" ),  
    "Is_Weekend", IF ( WEEKDAY ( [Date], 2 ) > 5, 1, 0 ),  
    "Week_of_Month", 1 + WEEKNUM ( [Date], 2 ) - WEEKNUM ( DATE ( YEAR ( [Date] ), MONTH ( [Date] ), 1 ), 2 ),  
    "Week_of_Month_Name", "Week " & ( 1 + WEEKNUM ( [Date], 2 ) - WEEKNUM ( DATE ( YEAR ( [Date] ), MONTH ( [Date] ), 1 ), 2 ) )  
)
```

This DAX code generates a dynamic calendar table spanning from 2024 to 2025 using CALENDAR and expands it with structured time intelligence attributes. It enriches the dates with custom year, quarter, month, and week dimensions using functions like ADDCOLUMNS and FORMAT to support seamless multi-period financial analysis.

Key DAX Calculations

```
User_Security =
DATATABLE (
    "UserEmail", STRING,
    "Role", STRING,
    "Company Code", STRING,
    "Allowed Department", STRING,
    {
        -----
        -- 997 - Centidel Group (centidelgroup.com)
        -----
        { "it.manager@centidelgroup.com", "IT Manager", "997", "IT & Technology" },
        { "sales.manager@centidelgroup.com", "Sales Manager", "997", "Sales & Marketing" },
        { "ops.manager@centidelgroup.com", "Operations Manager", "997", "Operations & Finance" },
        { "cfo@centidelgroup.com", "CFO", "997", "ALL" },

        -----
        -- 998 - Skyndu Solutions (skyndusolutions.com)
        -----
        { "it.manager@skyndusolutions.com", "IT Manager", "998", "IT & Technology" },
        { "sales.manager@skyndusolutions.com", "Sales Manager", "998", "Sales & Marketing" },
        { "ops.manager@skyndusolutions.com", "Operations Manager", "998", "Operations & Finance" },
        { "cfo@skyndusolutions.com", "CFO", "998", "ALL" },

        -----
        -- 999 - Cogilith Group (cogilithgroup.com)
        -----
        { "it.manager@cogilithgroup.com", "IT Manager", "999", "IT & Technology" },
        { "sales.manager@cogilithgroup.com", "Sales Manager", "999", "Sales & Marketing" },
        { "ops.manager@cogilithgroup.com", "Operations Manager", "999", "Operations & Finance" },
        { "cfo@cogilithgroup.com", "CFO", "999", "ALL" },

        -----
        -- CEO
        -----
        { "ceo@ceskcoholding.com", "Group CEO", "ALL", "ALL" }
    }
)
```

This DAX snippet creates a static DATATABLE to establish a Row-Level Security (RLS) mapping matrix based on user emails, corporate roles, company codes, and department permissions. It enforces granular, role-based data access control across subsidiary entities while granting unrestricted enterprise-wide visibility to executive leadership.

Key DAX Calculations

D

```
CAPEX_Actual =  
CALCULATE(  
    SUM('Transactional Data Table'[Actual_Amount]),  
    TREATAS(  
        VALUES('CAPEX Master Data Table'[Account Number]),  
        'Transactional Data Table'[Account Number]  
    )  
)
```

To keep the model lightweight and avoid schema clutter or circular dependencies, I used TREATAS to build a dynamic virtual relationship instead of a hard physical join.

First, VALUES pulls a clean, distinct list of valid CAPEX accounts from the master table.

TREATAS takes those accounts and maps them straight onto the transaction table on the fly, applying them as a targeted filter.

CALCULATE enforces that context so the engine focuses strictly on CAPEX-related line items, and SUM adds up the actual amounts. It keeps the model fast and flexible while delivering exact CAPEX numbers for executive reporting.

Key DAX Calculations

```
DSO (Days Sales Outstanding) =  
VAR _SelectedMinDate = MIN ( 'Date Conversion Table'[Date] )  
VAR _SelectedMaxDate = MAX ( 'Date Conversion Table'[Date] )  
  
VAR _DaysInPeriod = DATEDIFF ( _SelectedMinDate, _SelectedMaxDate, DAY ) + 1  
  
VAR _TotalAR =  
    CALCULATE (  
        SUM ( 'Transactional Data Table'[Actual_Amount] ),  
        'GL Account Master Data Table'[GL Category] = "Receivables",  
        DATESBETWEEN ( 'Date Conversion Table'[Date], _SelectedMinDate, _SelectedMaxDate )  
    )  
  
VAR _TotalSales =  
    CALCULATE (  
        SUM ( 'Transactional Data Table'[Actual_Amount] ),  
        'GL Account Master Data Table'[GL Category] = "Sales",  
        DATESBETWEEN ( 'Date Conversion Table'[Date], _SelectedMinDate, _SelectedMaxDate )  
    )  
  
RETURN  
    DIVIDE (  
        ABS ( _TotalAR ),  
        _TotalSales,  
        0  
    ) * _DaysInPeriod  
,
```

The DSO calculation dynamically captures the active reporting window via MIN and MAX date boundaries to evaluate the exact duration of the selected period. This guarantees accurate and responsive Days Sales Outstanding metrics across any custom slicer selection or user-filtered timeframe.



THANK YOU